

Danh mục: medication

Type 2 diabetes in adults: management

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NICE guideline

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Insulin-based treatments

1.32 Starting an insulin-based treatment

1.32.1

Provide a structured education programme to adults with type 2 diabetes starting insulin therapy. The programme should include:

injection technique, including rotating injection sites and avoiding repeated injections at the same point within sites

self-monitoring

dose titration to target levels

dietary advice

DVLA's Assessing fitness to drive: a guide for medical professionals

managing hypoglycaemia

managing acute changes in plasma glucose

support from a healthcare professional trained in insulin therapy.

[2015, amended 2026]

1.32.2

When initiating insulin for adults with type 2 diabetes:

continue to offer metformin to people already taking it

stop any other medicines being used solely to manage hyperglycaemia

discuss with the person the risks and benefits of continuing medicines for other benefits such as cardiovascular protection or weight management.

[2015, amended 2026]

Why the committee made these recommendations

In 2026, the insulin-based treatment recommendations were amended to reflect the withdrawal of insulin products and known insulin brand shortages. Based on the committee's clinical experience and consensus, this refresh acknowledges the increased use of analogue insulin. The committee agreed that:

different insulin therapies may be more useful for different people dependent on their symptoms (for example, if there is a risk of nocturnal hypoglycaemia, a longer acting basal insulin might be more suitable) and

the added flexibility of recommending broad drug classes rather than specific insulins will support people with diabetes and healthcare professionals to choose the most suitable treatment.

How the recommendations might affect practice

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The effect of the 2026 recommendation updates on current practice is uncertain. There may be a general decrease in insulin use because of the use of other medications.

1.33 Choosing a type of insulin

1.33.1

Offer basal insulin intended for administration once or twice a day to adults with type 2 diabetes as initial insulin therapy.

[2015, amended 2026]

1.33.2

As initial insulin therapy for adults with type 2 diabetes, especially if the person's HbA1c is 75 mmol/mol (9.0%) or higher, consider combining:

basal insulin intended for administration once or twice a day and

short or rapid acting insulin.

This should be injected either separately or as a pre-mixed (biphasic) insulin preparation.

[2015, amended 2026]

Why the committee made these recommendations

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How the recommendations might affect practice

The effect of the 2026 recommendation updates on current practice is uncertain. There may be a general decrease in insulin use because of the use of other medications.

1.34 Choosing a preparation

1.34.1

Make a shared decision with the person on the choice of basal insulin preparation, based on considerations that are specific to them, including whether:

the person needs help from a carer or healthcare professional to inject insulin or

there is a particular concern about nocturnal hypoglycaemia or

the person has a strong preference for once-daily injections.

When multiple basal insulin types (including biosimilars) and regimens are equally suitable for the person's needs, use the least expensive option.

[2015, amended 2026]

1.34.2

Consider pre-mixed preparations that include insulin analogues rather than including human insulin, if:

the person prefers injecting insulin immediately before a meal or

hypoglycaemia is a problem or

blood glucose levels rise markedly after meals.

[2015, amended 2026]

Why the committee made these recommendations

In 2026, the insulin-based treatment recommendations were amended to reflect the withdrawal of insulin products and

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known insulin brand shortages. Based on the committee's clinical experience and consensus, this refresh acknowledges the increased use of analogue insulin. The committee agreed that:

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the added flexibility of recommending broad drug classes rather than specific insulins will support people with diabetes and healthcare professionals to choose the most suitable treatment.

How the recommendations might affect practice

The effect of the 2026 recommendation updates on current practice is uncertain. There may be a general decrease in insulin use because of the use of other medications.

1.35 Reviews

1.35.1

At each review, check whether adults with type 2 diabetes who are on a basal insulin regimen also need a short or rapid acting bolus insulin before meals (or a pre-mixed [biphasic] insulin preparation).

[2015, amended 2026]

1.35.2

At each review, check whether adults with type 2 diabetes who are using a pre-mixed (biphasic) preparation and whose individualised

glycaemic targets

are not met, need to change to:

a different pre-mixed (biphasic) insulin preparation or

a basal-bolus regimen with basal insulin intended for administration once or twice a day.

[2015, amended 2026]

Why the committee made these recommendations

In 2026, the insulin-based treatment recommendations were amended to reflect the withdrawal of insulin products and known insulin brand shortages. Based on the committee's clinical experience and consensus, this refresh acknowledges the increased use of analogue insulin. The committee agreed that:

different insulin therapies may be more useful for different people dependent on their symptoms (for example, if there is a risk of nocturnal hypoglycaemia, a longer acting basal insulin might be more suitable) and

the added flexibility of recommending broad drug classes rather than specific insulins will support people with diabetes and healthcare professionals to choose the most suitable treatment.

How the recommendations might affect practice

The effect of the 2026 recommendation updates on current practice is uncertain. There may be a general decrease in insulin use because of the use of other medications.

1.35.3

Follow the

MHRA's guidance on minimising the risk of medication error with high strength, fixed combination and biosimilar insulin products

, including its advice for healthcare professionals when starting treatment with a biosimilar.

[2021]

1.35.4

When people are already using an insulin for which a lower cost biosimilar is available:

Discuss with the person the possibility of switching to the biosimilar.

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Make a shared decision about it with them.

[2021]

Why the committee made these recommendations

Biosimilars have the potential to offer the NHS considerable cost savings. To gain approval for use, biosimilar medicines have to be shown to be safe and as effective as the original reference medicine, and to have the same quality. Based on this, the committee noted it was appropriate, when starting a new prescription of an insulin for which a biosimilar is available, to use the one with the lowest cost.

In addition, people may be using an insulin for which a lower cost biosimilar is available. In such cases, the committee recommended discussing with people the possibility of switching to the biosimilar. This could happen at the person's routine review. They also agreed that switching to the biosimilar should be carefully planned. A shared decision should be reached about the switch, taking into consideration the dose-switching protocols, monitoring, and the person's concerns about switching from their existing regimen. Healthcare professionals should also refer to the summary of product characteristics for further information when considering switching to biosimilars.

1.36 Insulin delivery

1.36.1

For guidance on insulin delivery for adults with type 2 diabetes, see the section on insulin injection delivery in NICE's guideline on type 1 diabetes

[2015]